STEM Notes

Revision Notes

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2026-08-07

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Chapter 1
Mathematics

1.1 Partial Derivatives

Definition 1.1.1

The partial derivative of $f : \mathbb{R}^n \to \mathbb{R}$ with respect to x_i at (x) is

$$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f((x) + he_i) - f((x))}{h}.$$

Lemma 1.1.1 Descent Direction

If $\nabla f \neq 0$, then $-\nabla f$ is a descent direction:

$$\exists \alpha > 0, f((x) - \alpha \nabla f) < f((x)).$$

Proof. By Taylor:

$$f((x) - \alpha \nabla f) = f((x)) - \alpha \left|\nabla f\right|^2 + \mathcal{O}(\alpha^2).$$

For small α, the linear term dominates. ■ ■

Theorem 1.1.1 Chain Rule

If $f(\boldsymbol{x})$ and $\boldsymbol{x}(\boldsymbol{u})$ are differentiable, then

$$\frac{\partial f}{\partial u_i} = \sum_j \frac{\partial f}{\partial x_j} \frac{\partial x_j}{\partial u_i}.$$

1.2 Common Gradients

f	∇f	Convex?	Note
$\left\|x\right\|^2$	$2x$	Yes	Strongly
$x^T A x$	$(A + A^T)x$	if $A \succeq 0$	Quadratic
$-\log x$	$-\frac{1}{x}$	Yes	Log barrier
e^x	e^x	Yes	Strictly

Chapter 2
Physics

2.1 Maxwell's Equations

Property 2.1.1 Gauss's Law

$$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0},$$

where $\boldsymbol{E}$ is the electric field.

Property 2.1.2 Faraday's Law

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}.$$

A time-varying $\boldsymbol{B}$ induces a circulating $\boldsymbol{E}$.

Property 2.1.3 Ampère-Maxwell Law

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}.$$

Law	Integral Form	Differential Form
Gauss	$\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A} = \frac{q_{\text{enc}}}{\varepsilon_0}$	$\nabla \cdot \boldsymbol{E} = \frac{\rho}{\varepsilon_0}$
Faraday	$\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{l} = -\frac{\mathrm{d}}{\mathrm{d}t} \int \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{A}$	$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$
Ampère	$\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{\mathrm{d}}{\mathrm{d}t} \int \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A}$	$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}$

2.2 Electromagnetic Wave

Definition 2.2.1

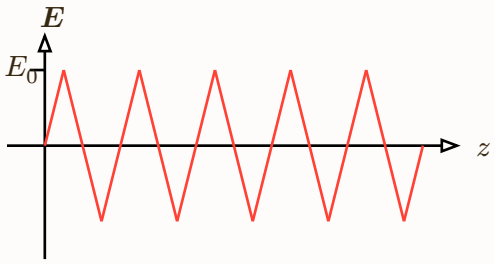
In free space, taking curl of Faraday and substituting Ampère:

$$\nabla^2 \boldsymbol{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2}.$$

This is the wave equation with speed $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.

Example 2.2.1 Plane Wave

$\boldsymbol{E}(z,t) = E_0 \cos(kz - \omega t)\hat{\boldsymbol{x}}$ propagates along z.



Wavelength $\lambda = 2\frac{\pi}{k}$, visible light: $(380 - 750)\,\mathrm{nm}$.[1]

Problem 2.1

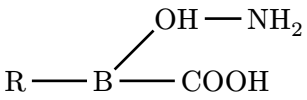
Show $\boldsymbol{B}(z,t) = \left(\frac{E_0}{c}\right)\cos(kz - \omega t)\hat{\boldsymbol{y}}$ satisfies Faraday's law.

Solution. $\nabla \times \boldsymbol{E} = \frac{\partial E_x}{\partial z}\hat{\boldsymbol{y}} = -kE_0 \sin(kz - \omega t)\hat{\boldsymbol{y}} = -\frac{\partial \boldsymbol{B}}{\partial t}$. ■ □

[1] $c \approx 2.998 \cdot 10^8\,\mathrm{m\,s^{-1}}$.

Chapter 3
Chemistry

3.1 Chemical Structures



3.2 Key Reactions

Reaction	Type	ΔH
$\mathrm{N_2 + 3H_2 \rightarrow 2NH_3}$	Haber	$-92.4\,\mathrm{kJ\,mol^{-1}}$
$\mathrm{C + O_2 \rightarrow CO_2}$	Combustion	$-393.5\,\mathrm{kJ\,mol^{-1}}$
$\mathrm{H_2O \rightarrow H^+ + OH^-}$	Dissociation	$55.8\,\mathrm{kJ\,mol^{-1}}$

> **Remark 3.2.1**
>
> Reaction rates follow Arrhenius: $k = A e^{-\frac{E_a}{RT}}$, where E_a is activation energy (typically $(40 - 200)\,\mathrm{kJ\,mol^{-1}}$).

Chapter 4
Computer Science

4.1 Gradient Descent

```
GRADIENT DESCENT():
Input: f : ℝ^n → ℝ, x_0, step α > 0, tolerance ε
k := 0
while |∇f(x_k)| > ε do
                                        // Descent direction
    d_k := −∇f(x_k)
                                        // Line search
    x_{k+1} := x_k + α d_k
    k := k + 1
    return x_k
```

Theorem 4.1.1 Convergence

For L-smooth, μ-strongly convex f with $\alpha = \frac{2}{L+\mu}$:

$$|x_k - x^*| \leq \left(\frac{\kappa - 1}{\kappa + 1}\right)^k |x_0 - x^*|,$$

where $\kappa = \frac{L}{\mu}$.

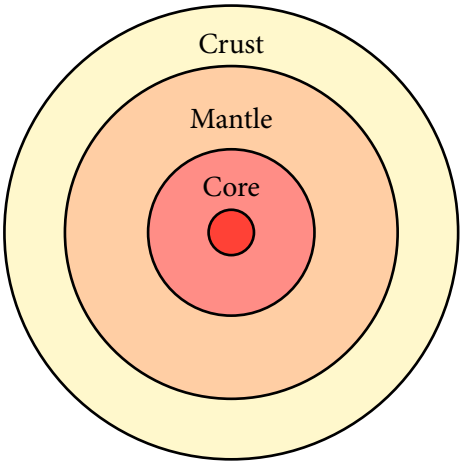
Problem 4.1

Implement GD for $f(x) = (x-3)^2$ with $x_0 = 0$, $\alpha = 0.1$. How many iterations to reach $|f'(x)| < 10^{-3}$?

Solution. $f'(x) = 2(x-3)$, $x_{k+1} = x_k - 0.2(x_k - 3) = 0.8x_k + 0.6$. Starting from 0: $x_1 = 0.6$, $x_2 = 1.08$, … converges to 3. Error halves roughly every 3 iterations. ■ □

Chapter 5
Earth Science

5.1 Earth's Structure



5.2 Key Properties

Layer	Depth	Temp	State
Crust	$(0-35)$ km	$(0-400)$	Solid
Mantle	$(35-2890)$ km	$(1000-3700)$	Solid
Outer Core	$(2890-5150)$ km	$(3700-4300)$	Liquid
Inner Core	$(5150-6371)$ km	$(4300-5500)$	Solid

Property 5.2.1 Seismic Waves

P-waves travel through solids and liquids at $v_P = \sqrt{\frac{K+\frac{4}{3}G}{\rho}}$, while S-waves only through solids at $v_S = \sqrt{\frac{G}{\rho}}$.

Remark 5.2.1

The Earth's magnetic field ($\approx 50\,\mu\text{T}$ at the surface) is generated by convection in the liquid outer core (geodynamo).[2]

[2] Earth's mass: $5.97 \cdot 10^{24}$ kg, radius: 6371 km.